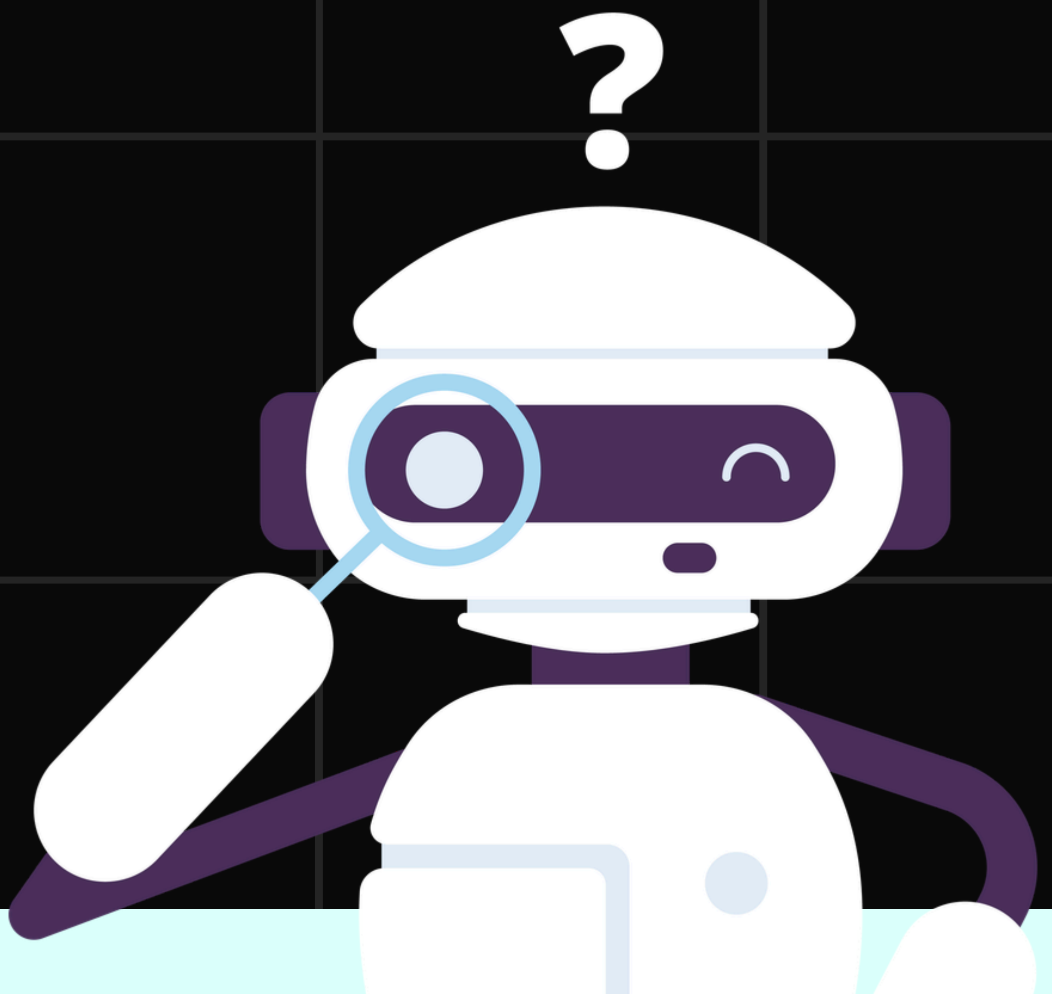
NALAPRO

20 NEWSGROUP

Presented by **Talha Murtaza**

STEP 0

DATA PREPROCESSING

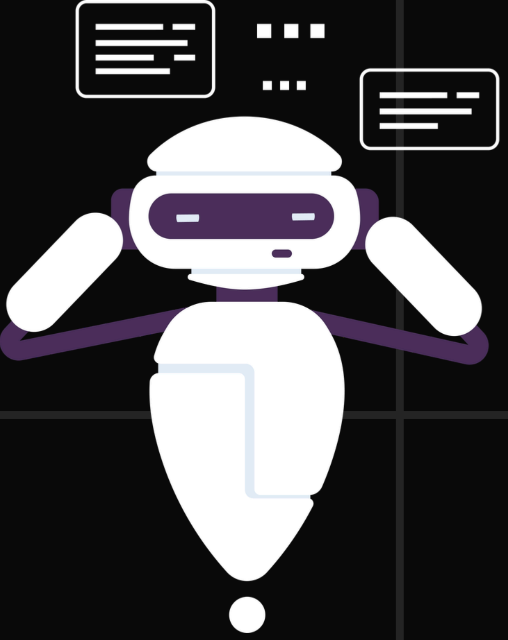


Pre-Processing Steps

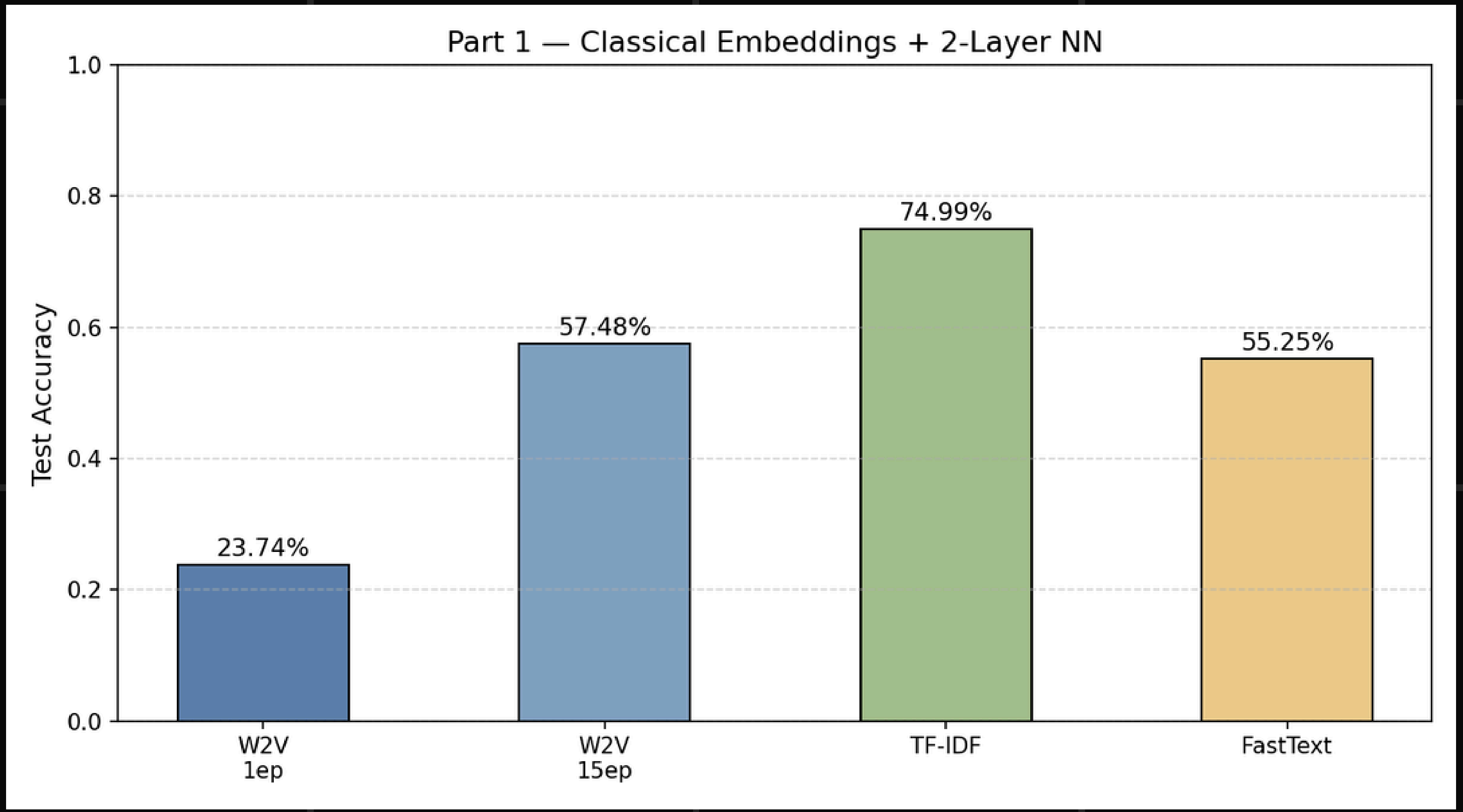
Lowercase Conversion - Regex Cleaning - Tokenization -

Train/Test Split & Defining Metrics

Train/Test Split for fair evaluation and defining all parameters at the start of the experiment



STEP 1



Neural Network

- 2 Layers with ReLU
- 10 Epochs
- Dropout = 0.3
- Cross Entropy Loss Function
- Adam Optimizer

TF-IDF

- Clean Input (15 Epochs)
- Max Feature: 10,000 - Min: 2
- Trained Neural Network for 10 Epochs on TF-IDF Data

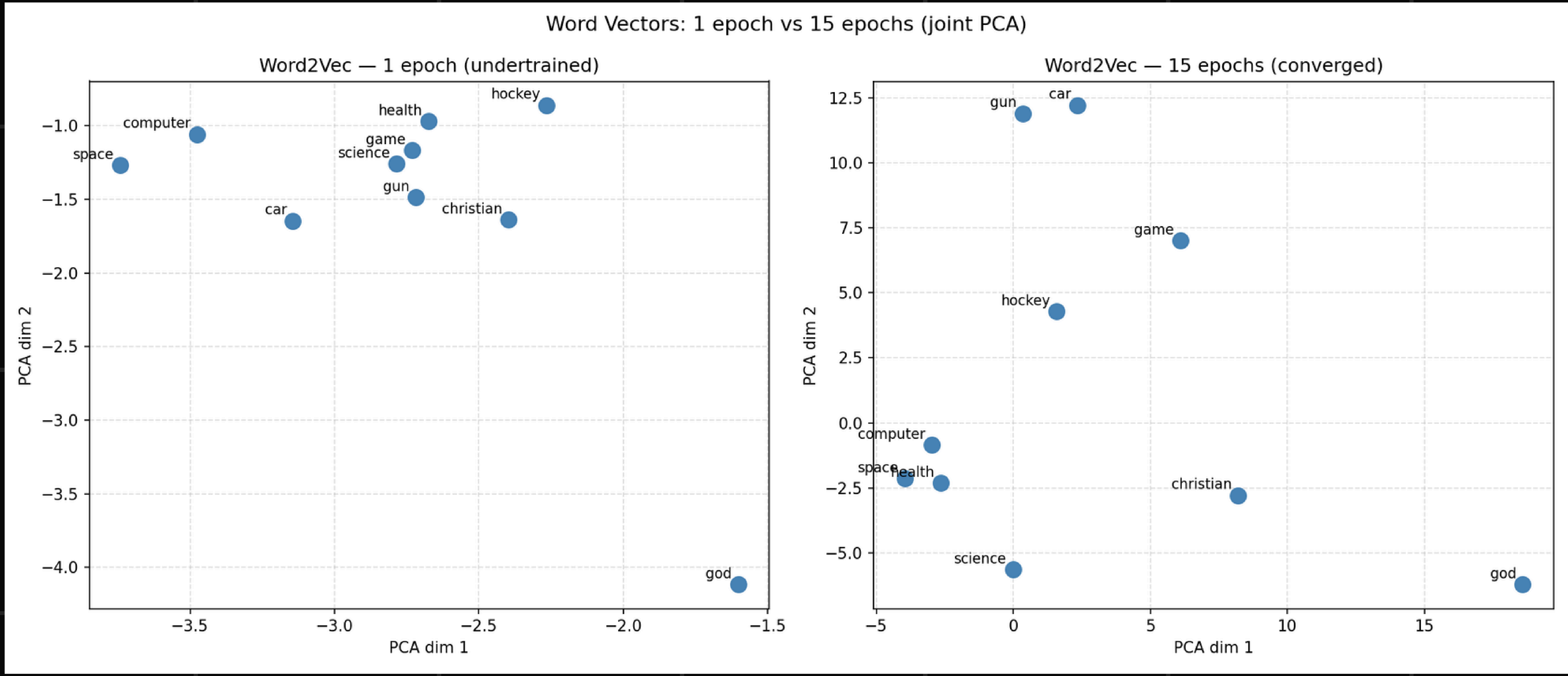
Word2Vec

- 1 vs 15 Epochs
- Tokenized Input - V-Size: 100
- PCA & Euclidean Distance to display difference
- Trained Neural Network for 10 Epochs on Word2Vec Data

FastText

- 15 Epochs - V-Size: 100 (Same as Word2Vec)
- Trained Neural Network for 10 Epochs on Fast-Text Data

WORD2VEC


Word Vectors: 1 epoch vs 15 epochs (joint PCA)
Word2Vec — 1 epoch (undertrained)
PCA dim 1
PCA dim 2
computer
space
health
hockey
game
science
gun
car
christian
god
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
−3.5
−3.0
−2.5
−2.0
−1.5
Word2Vec — 15 epochs (converged)
PCA dim 1
PCA dim 2
gun
car
game
hockey
computer
space
health
christian
science
god
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−5
0
5
10
15

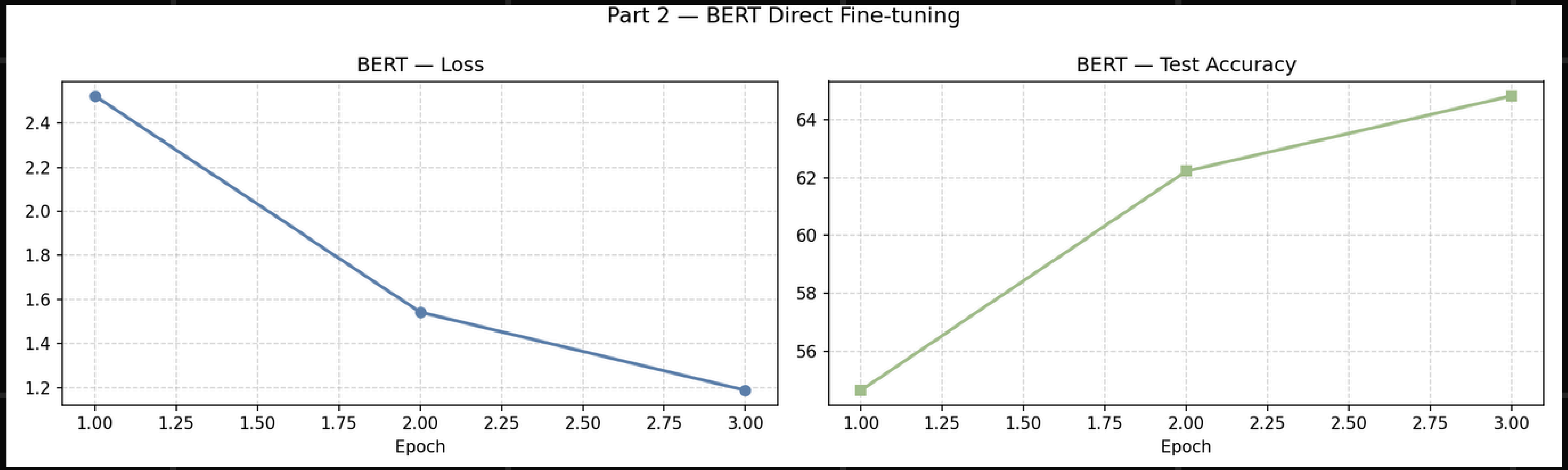
BERT

Tokenizer & Data Parameters

- Bert-base Uncased
- 3000 Documents Used
- Max Sequence Length: 128
- Batch Size: 16

BERT Fine-Tuning

- AdamW Optimizer
- Weight Decay: 0.01 - EPS = 1e-8
- Warmup Ratio: 0.1 - G-Clipping: 1.0
- Loss Function: nn.CrossEntropyLoss()



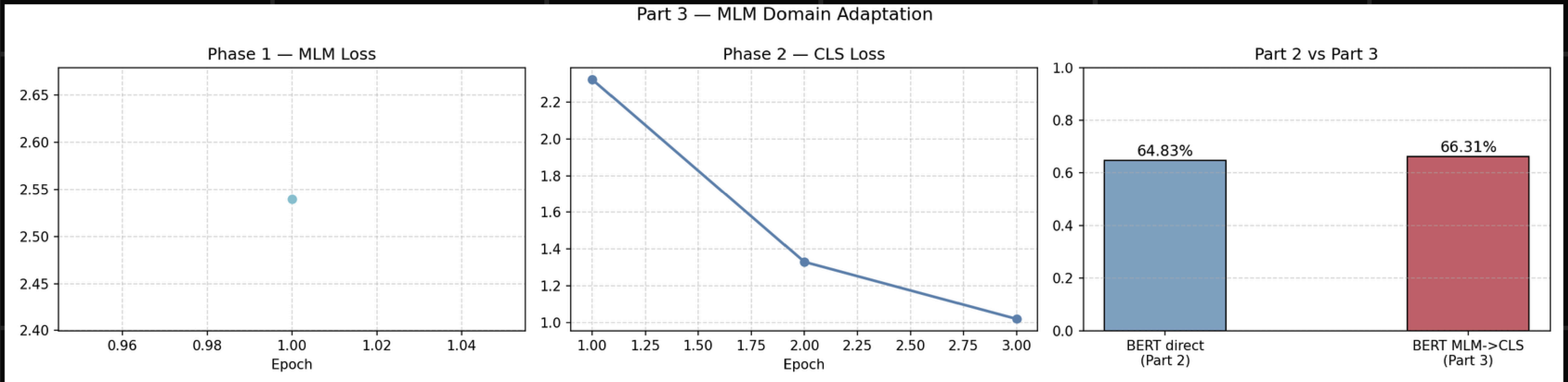
MLM ---> CLASSIFICATION

MLM

- Done on pretrained BERT
- Use of Data Collator
- Masked 15% of data
- 1 Epoch

MLM ---> CLASSIFICATION

- Done on MLM Trained BERT
- 3 Epochs
- Mostly Similar to BERT (Step 2)



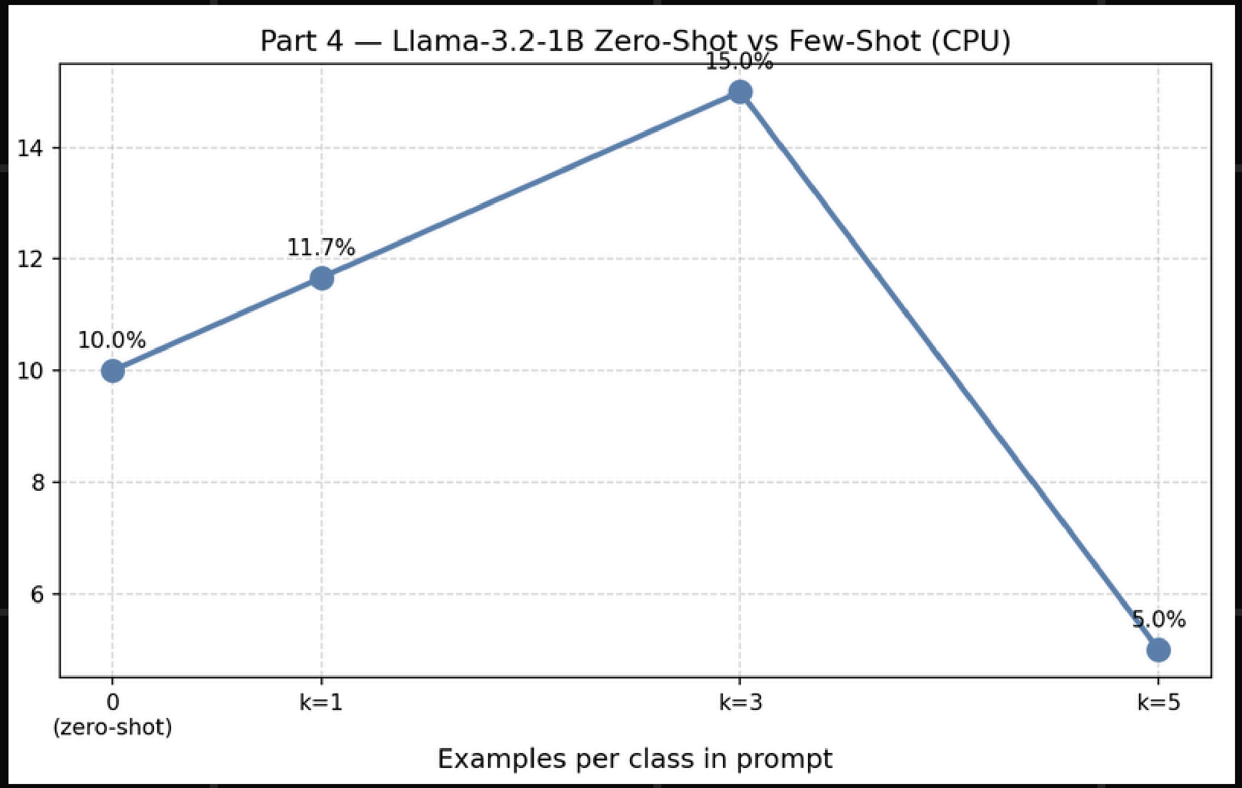
LLAMA3.2

LLAMA 3.2 (1B)

- Computational Limit
- Prompt Engineering
- Used for zero shot and few-shot classifications

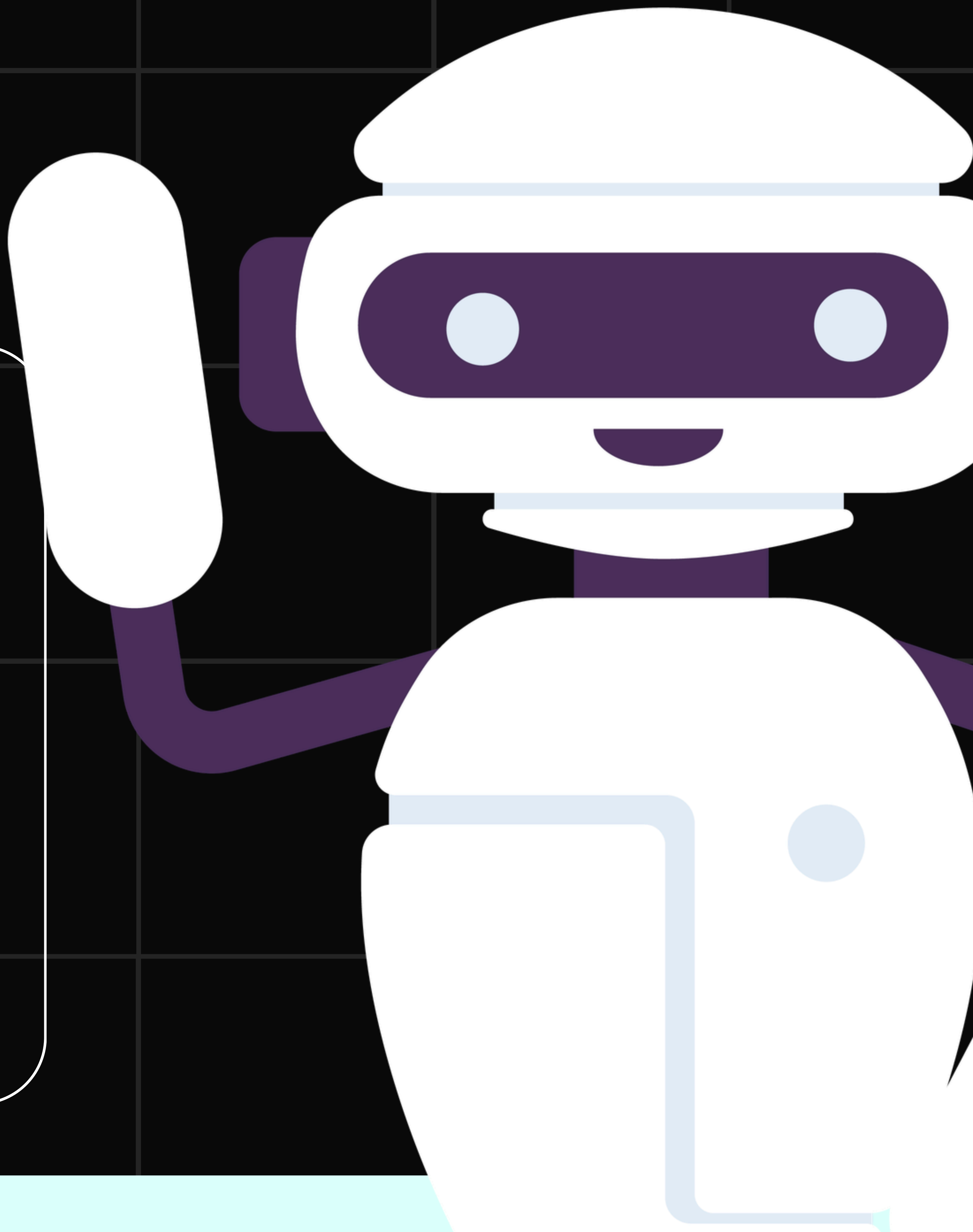
LLAMA 3.2 (1B)

- Used 20 Documents
- 60 for k=3
- 100 for k=5
- Context Window Exhaustion



FAILED ATTEMPTS

- Google GPU NVDIA
- K = 5
- Kaggle Notebook
- LLAMA3



FUTURE PLANS

BERT - LARGE

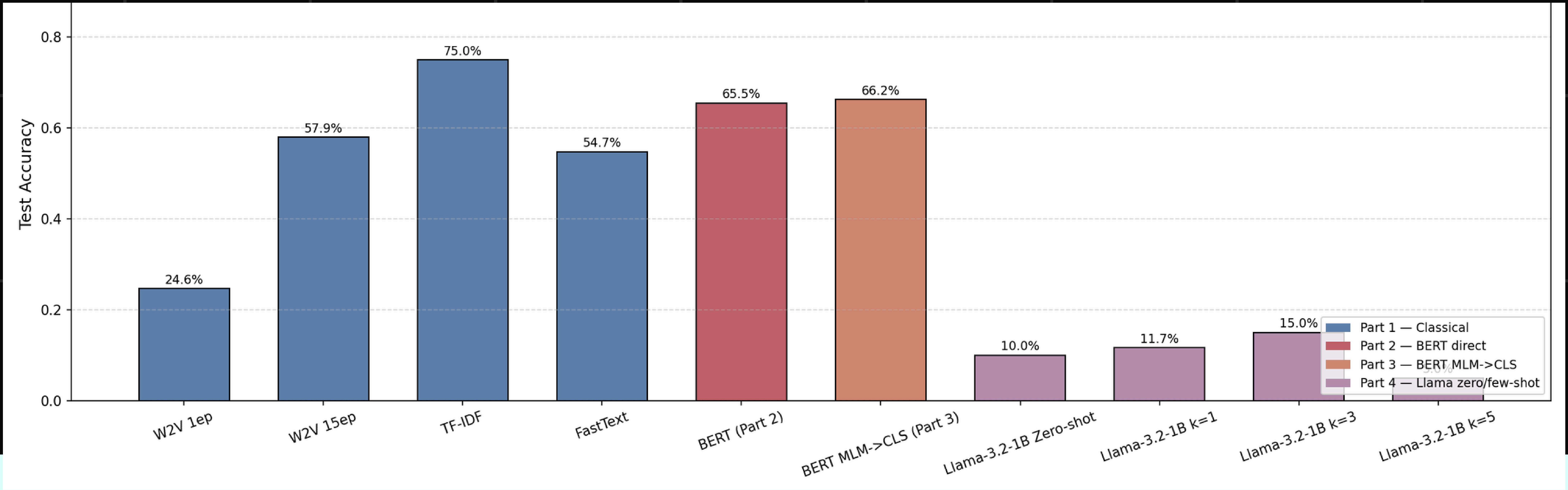
Optimizing with Full-BERT 64BIT with 15,000 Docs instead of 3000 Docs. Expected Accuracy: 82-90%.

MLM ---> CLASSIFICATION

Using 3000 Documents for MLM, 15000 can increase accuracy to 70-75%. At least 10 Epochs could move it further.

LLAMA

- Using LLAMA 3
- LoRA Fine-Tuning
- Using Full Dataset
- Accuracy Reduced because of LLAMA 3.2(1B)
- Expected Accuracy with LLAMA3 for each part is around 70-75%
- k = 5 would be an exception as it is throwing 100 documents as input but still will perform better using LLAMA3 (8B)



Thankyou